Lab 7: Fuzzing for Input Validation Bugs

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Assigned repo: SETAPESU26/L6_L7_15_movie_tickets (Lab 7: movie_tickets.py)

Tools: Python 3, Hypothesis, pytest

Step 1: Install dependencies

`pip install hypothesis` and `pip install pytest`

Step 2: Run the buggy application

Valid input works, but a seat with no number (H) crashes the program with a ValueError.

LAB_7 - python movie_tickets.py (buggy)

```
$ python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): H12
Row and Seat: ('H', 12)

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 150,250,400
Total with GST: 944.0

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3): 4 5 3
Average Rating: 4.0

$ python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): H
Traceback (most recent call last):
  File "movie_tickets.py", line 29, in <module>
    main()
    ~~~~^^
  File "movie_tickets.py", line 15, in main
    result = parse_seat(user_input)
  File "movie_tickets.py", line 2, in parse_seat
    return seat[0].upper(), int(seat[1:])
                            ~~~^^^^^^^^^^
ValueError: invalid literal for int() with base 10: ''
```

Step 3: Complete test_movie_tickets.py with Hypothesis

Nine property-based tests in three groups: (1) **no-crash** tests feeding `st.text() | st.none()` to each function; (2) **targeted** tests using `st.from_regex` to generate almost-valid inputs (empty fields, signs, spaces, Unicode digits, out-of-range ratings) and checking the result is sensible; (3) **correctness** tests on valid generated inputs, so the fixes cannot break normal behaviour.

Step 4: Run the tests against the buggy code

6 of 9 tests fail. Hypothesis shrank each failure to a minimal input (see table below).

```
LAB_7 - pytest on buggy code

$ python -m pytest test_movie_tickets.py -v --tb=no
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
hypothesis profile 'default'
rootdir: .../LAB_7
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 9 items

test_movie_tickets.py::test_parse_seat_no_crash FAILED                   [ 11%]
test_movie_tickets.py::test_total_price_no_crash FAILED                  [ 22%]
test_movie_tickets.py::test_average_rating_no_crash FAILED               [ 33%]
test_movie_tickets.py::test_parse_seat_seat_like_inputs FAILED           [ 44%]
test_movie_tickets.py::test_total_price_csv_like_inputs FAILED           [ 55%]
test_movie_tickets.py::test_average_rating_ratings_like_inputs FAILED    [ 66%]
test_movie_tickets.py::test_parse_seat_valid PASSED                      [ 77%]
test_movie_tickets.py::test_total_price_valid PASSED                     [ 88%]
test_movie_tickets.py::test_average_rating_valid PASSED                  [100%]

========================= short test summary info ==========================
FAILED test_movie_tickets.py::test_parse_seat_no_crash - ExceptionGroup: Hypo...
FAILED test_movie_tickets.py::test_total_price_no_crash - ExceptionGroup: Hyp...
FAILED test_movie_tickets.py::test_average_rating_no_crash - ExceptionGroup: ...
FAILED test_movie_tickets.py::test_parse_seat_seat_like_inputs - ExceptionGro...
FAILED test_movie_tickets.py::test_total_price_csv_like_inputs - ValueError: ...
FAILED test_movie_tickets.py::test_average_rating_ratings_like_inputs - Excep...
======================== 6 failed, 3 passed in 0.53s ========================
```

Function	Minimal failing input	Error raised
parse_seat	None	TypeError: not subscriptable
parse_seat	''	IndexError: string index out of range
parse_seat	'0' (or 'H')	ValueError: int('')
total_price	None	AttributeError: no .split
total_price	'' (or '1,,3')	ValueError: int('')
average_rating	None	AttributeError: no .split
average_rating	''	ZeroDivisionError
average_rating	':' / 'x'	ValueError: non-numeric rating
average_rating	'0'	Wrong result: 0.0 is outside the 1-5 rating range

Step 5: Fix the application

Each function now validates its input and returns `None` instead of raising; `main()` prints "Invalid input" in that case.

• **parse_seat**: rejects non-strings, strings shorter than 2 characters, a row that is not an ASCII letter (e.g. 'ß'.upper() is 'SS'), and a seat number that is not plain ASCII digits or is 0. Surrounding whitespace is stripped.

• **total_price**: rejects non-strings and any field that is empty, negative or non-numeric; whitespace around fields is allowed.

• **average_rating**: rejects non-strings, empty input (no more division by zero), non-numeric values and ratings outside 1-5.

```
LAB_7 - python movie_tickets.py (fixed)

$ python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): H12
Row and Seat: ('H', 12)

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 150,250,400
Total with GST: 944.0

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3): 4 5 3
Average Rating: 4.0

$ python movie_tickets.py
==== Running parse_seat ====
Enter a seat number (e.g. H12): H
Row and Seat: Invalid input

==== Running total_price ====
Enter ticket prices as CSV (e.g. 150,250,400): 150,,400
Total with GST: Invalid input

==== Running average_rating ====
Enter ratings separated by spaces (e.g. 4 5 3):
Average Rating: Invalid input
```

Step 6: Re-run the tests against the fixed code

All 9 tests pass.

LAB_7 - pytest on fixed code

```
$ python -m pytest test_movie_tickets.py -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
hypothesis profile 'default'
rootdir: .../LAB_7
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test_movie_tickets.py::test_average_rating_valid PASSED                  [100%]

============================== 9 passed in 2.48s ===============================
```